

Assignment No 16

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Domain : AI & ML

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Q1.

```
import pandas as pd

df = pd.read_csv("insurance.csv")
print(df.head(10))
```

...	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520
5	31	female	25.740	0	no	southeast	3756.62160
6	46	female	33.440	1	no	southeast	8240.58960
7	37	female	27.740	3	no	northwest	7281.50560
8	37	male	29.830	2	no	northeast	6406.41070
9	60	female	25.840	0	no	northwest	28923.13692

Q2.

```
shape of the dataset:
... (1338, 7)
Information about the dataset:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         1338 non-null   int64
1   sex         1338 non-null   object
2   bmi         1338 non-null   float64
3   children    1338 non-null   int64
4   smoker      1338 non-null   object
5   region      1338 non-null   object
6   charges     1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB

Statistical summary:
              age          bmi      children      charges
count  1338.000000  1338.000000  1338.000000  1338.000000
mean     39.207025   30.663397    1.094918  13270.422265

Statistical summary:
              age          bmi      children      charges
count  1338.000000  1338.000000  1338.000000  1338.000000
mean     39.207025   30.663397    1.094918  13270.422265
std      14.049960    6.098187    1.205493  12110.011237
min      18.000000   15.960000    0.000000   1121.873900
25%      27.000000   26.296250    0.000000   4740.287150
50%      39.000000   30.400000    1.000000   9382.033000
75%      51.000000   34.693750    2.000000  16639.912515
max      64.000000   53.130000    5.000000  63770.428010
```

Q3.

... Null values in each column:

age 0

sex 0

bmi 0

children 0

smoker 0

region 0

charges 0

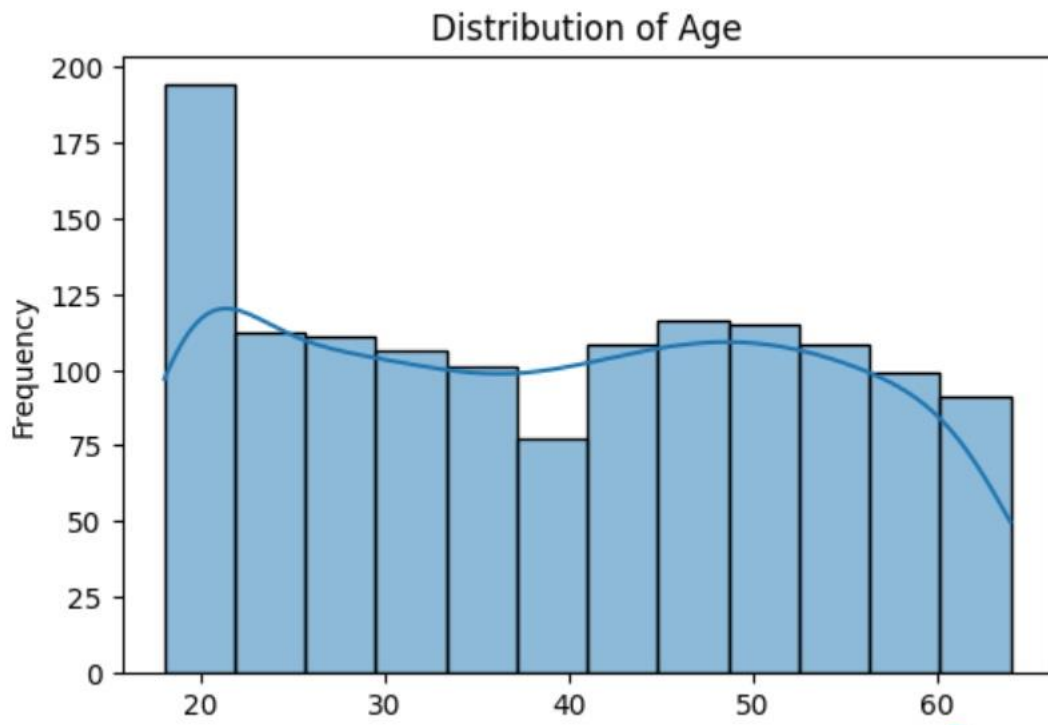
dtype: int64

Q4.

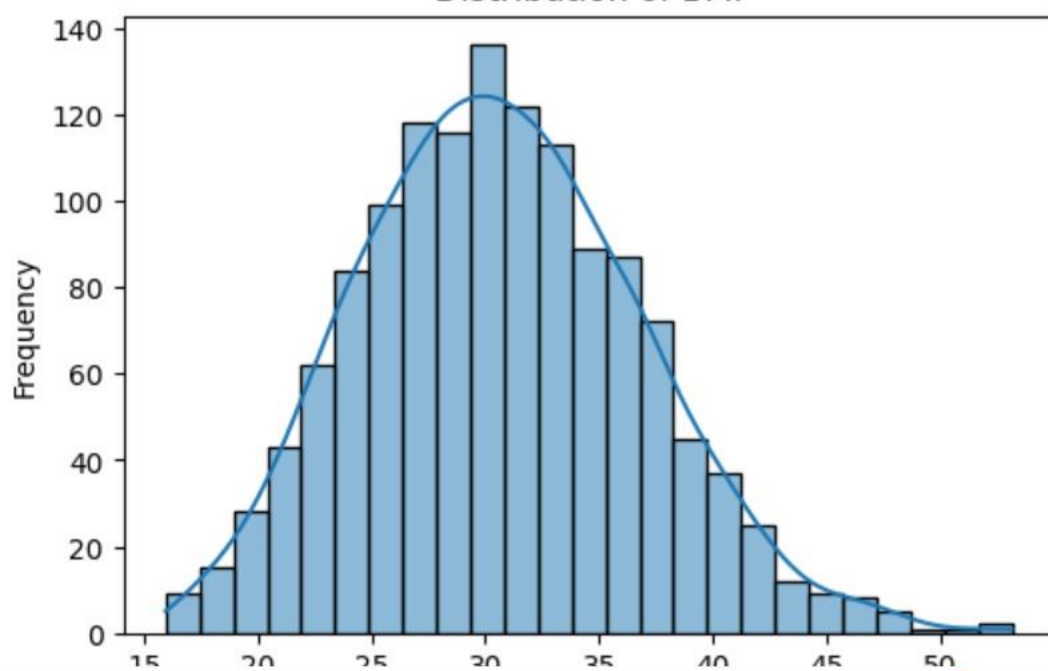
```
... Numerical Columns:  
Index(['age', 'bmi', 'children', 'charges'], dtype='object')  
  
Categorical Columns:  
Index(['sex', 'smoker', 'region'], dtype='object')
```

Q5.

...

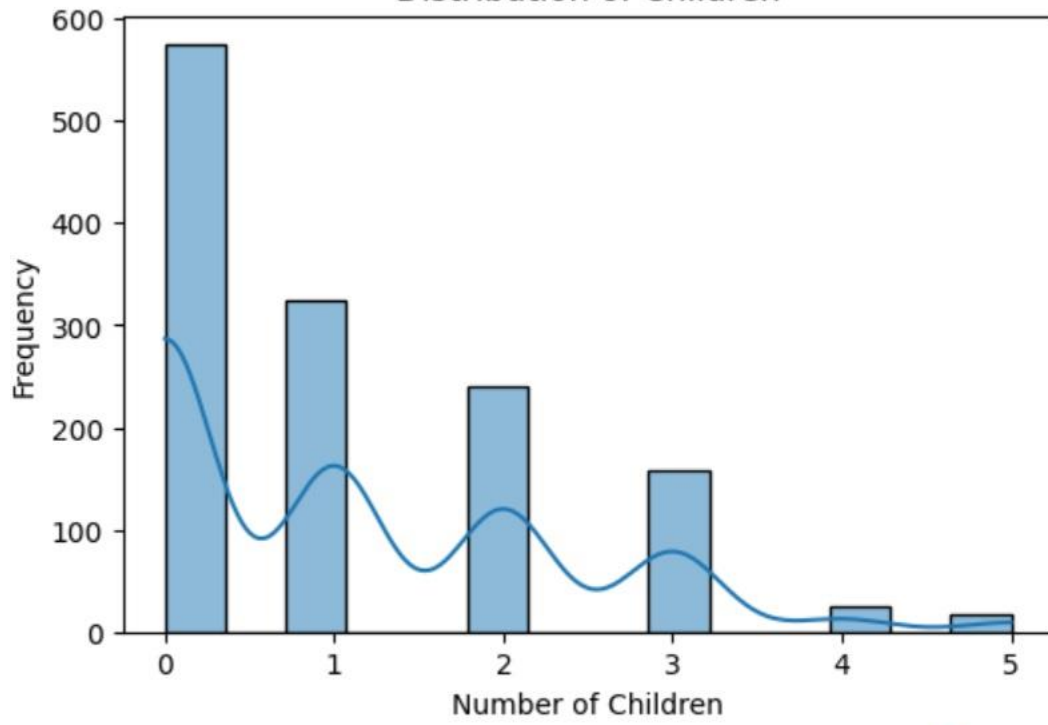


Distribution of BMI

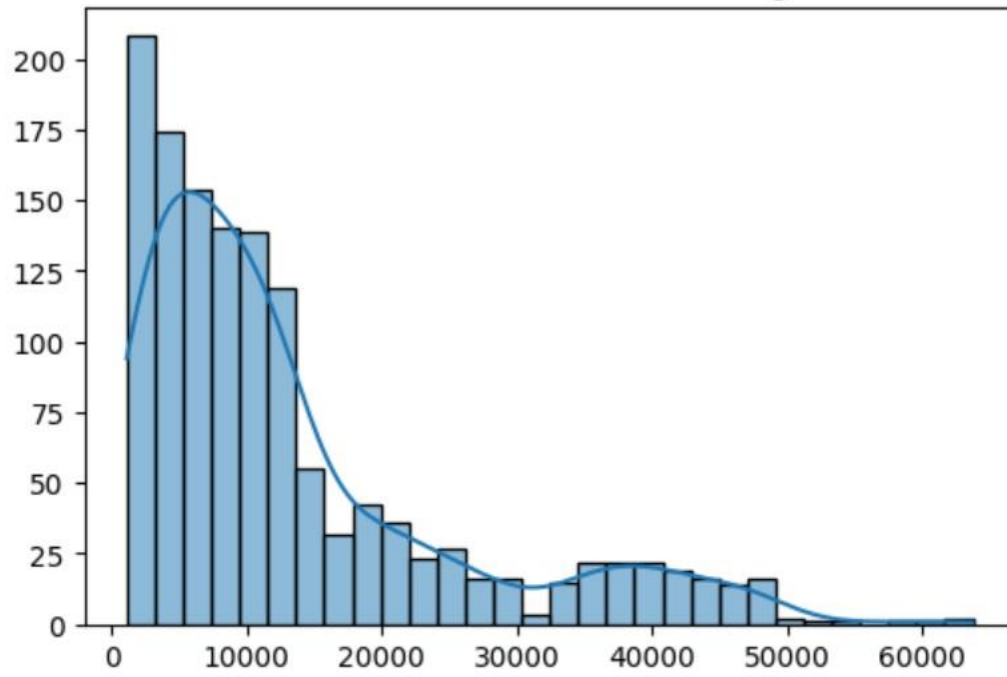


BMI

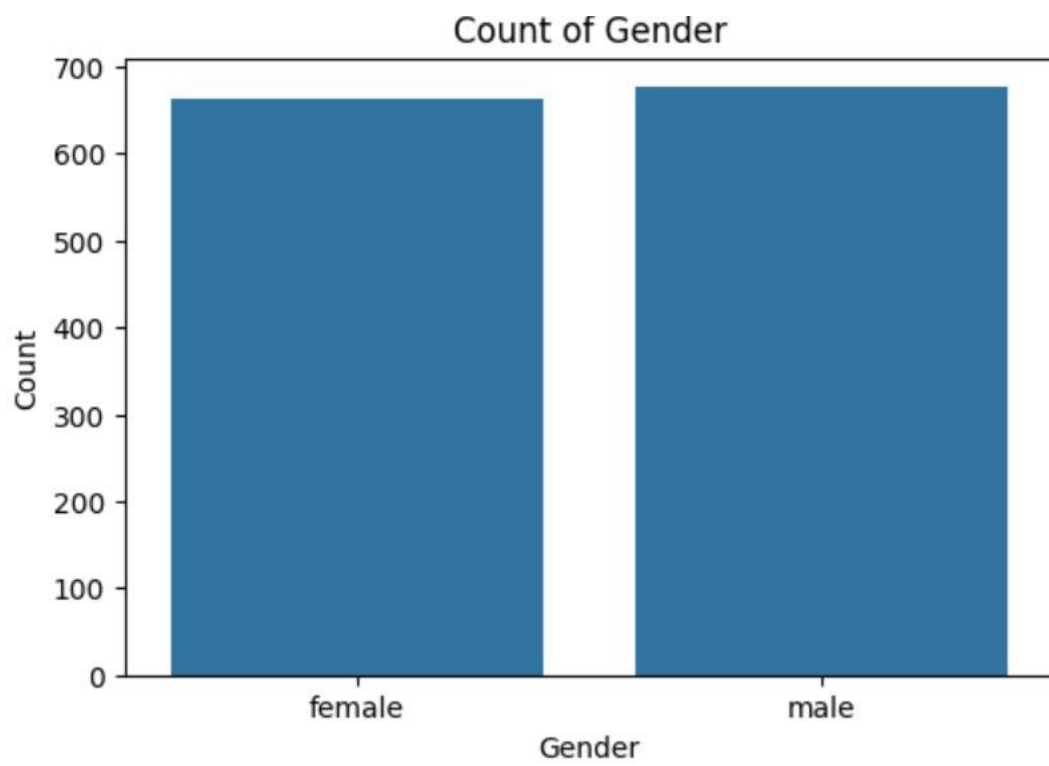
Distribution of Children

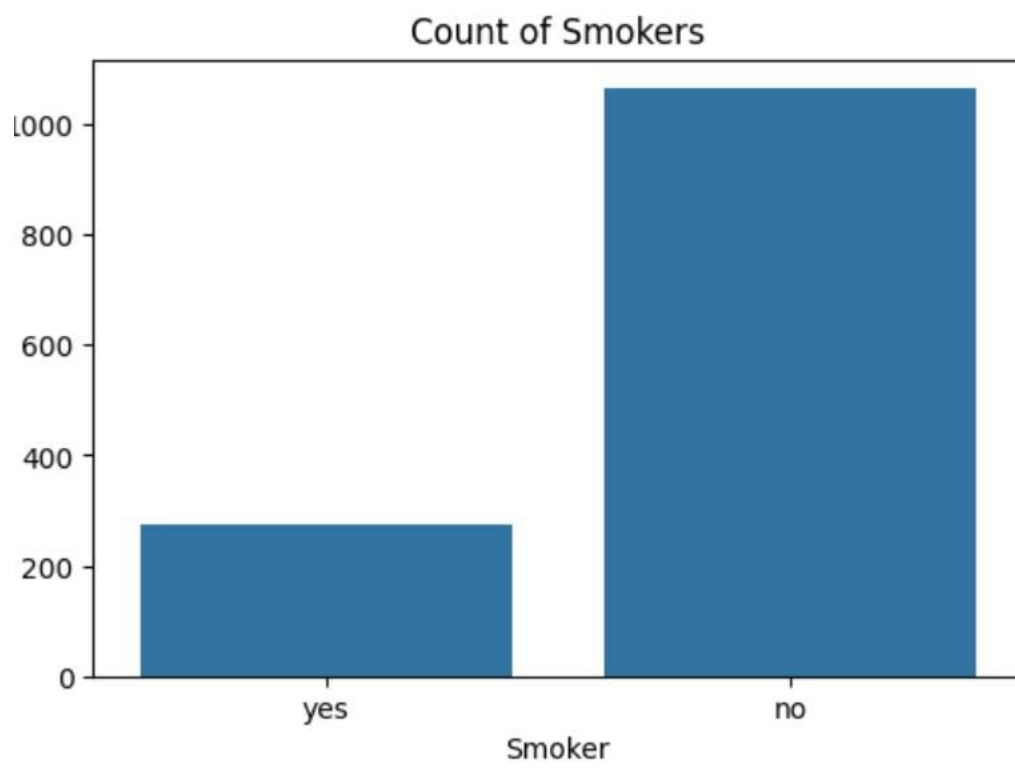


Distribution of Insurance Charges

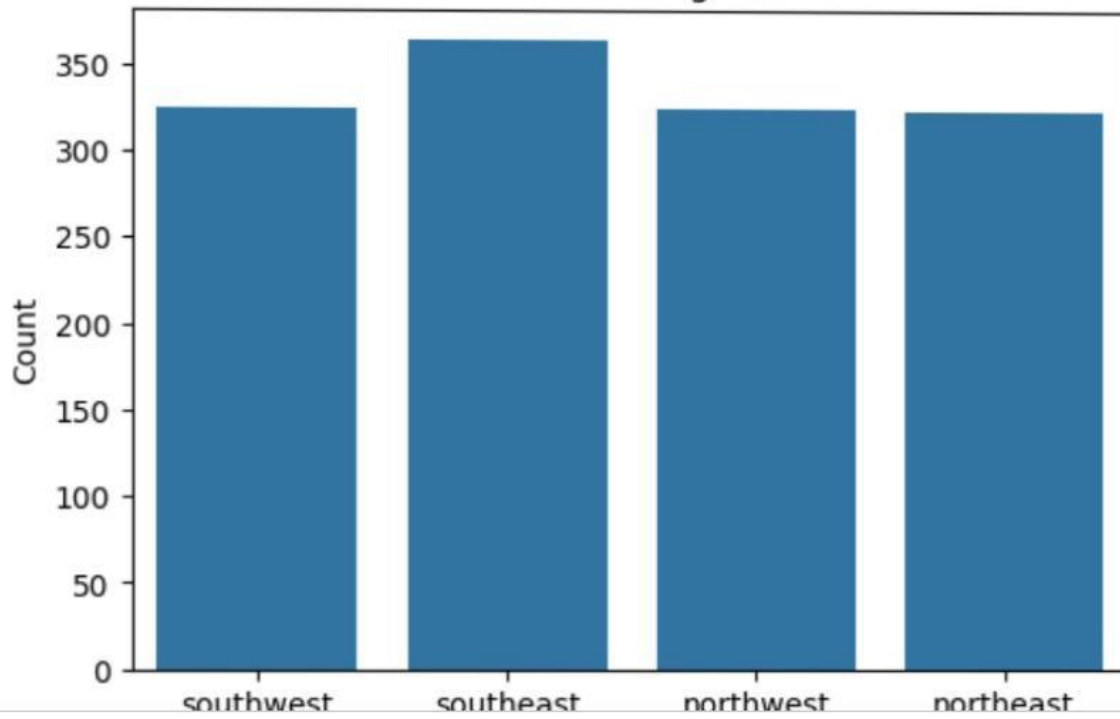


Q6.

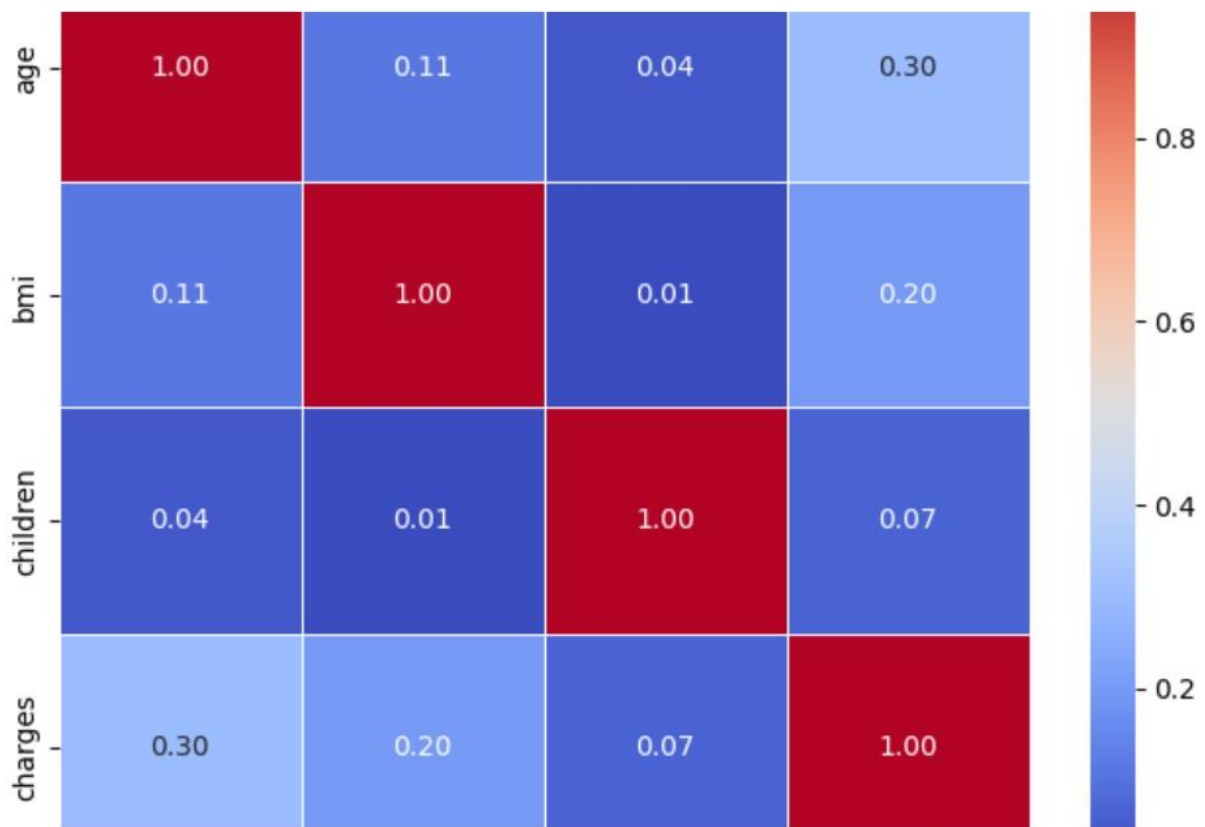




Count of Regions



Q7.



Q8.

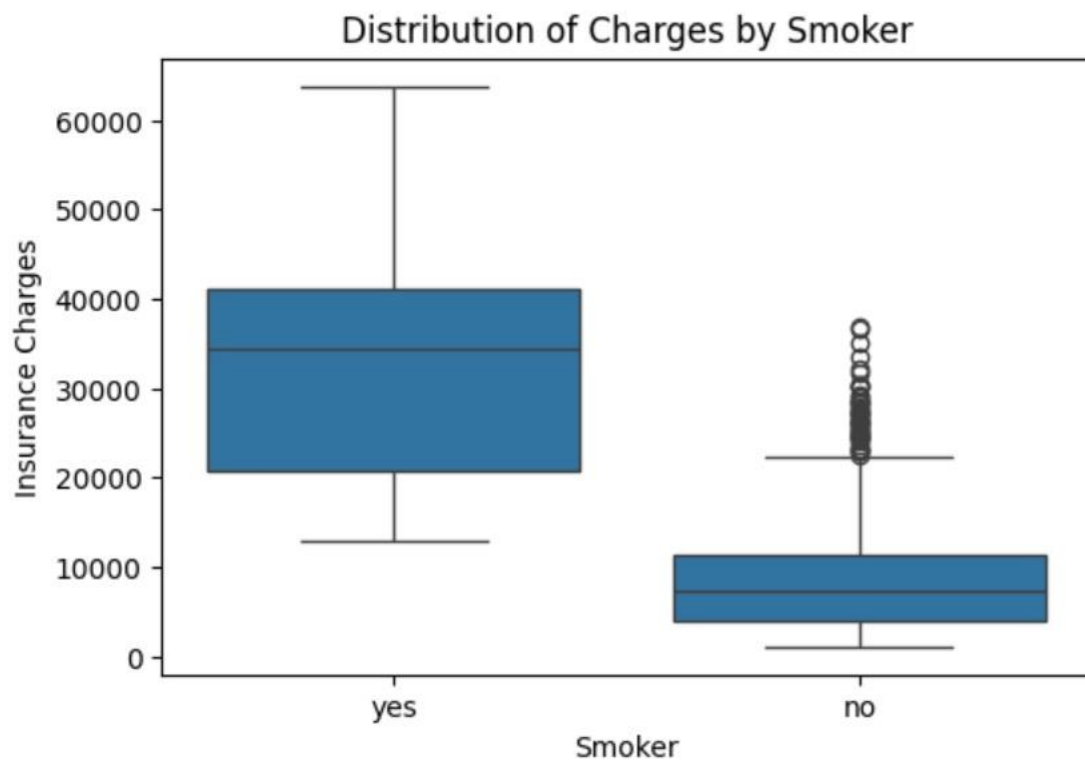
```
... Average Insurance Charges:
      13270.422265141257

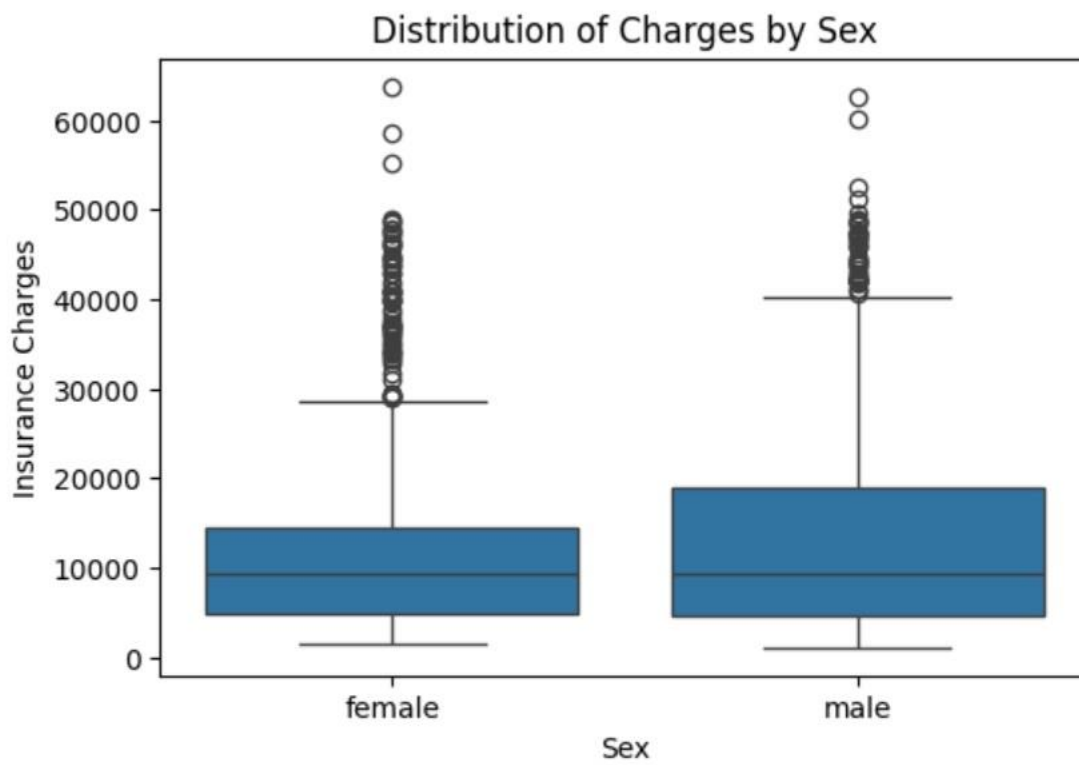
      Maximum Insurance Charges:
      63770.42801

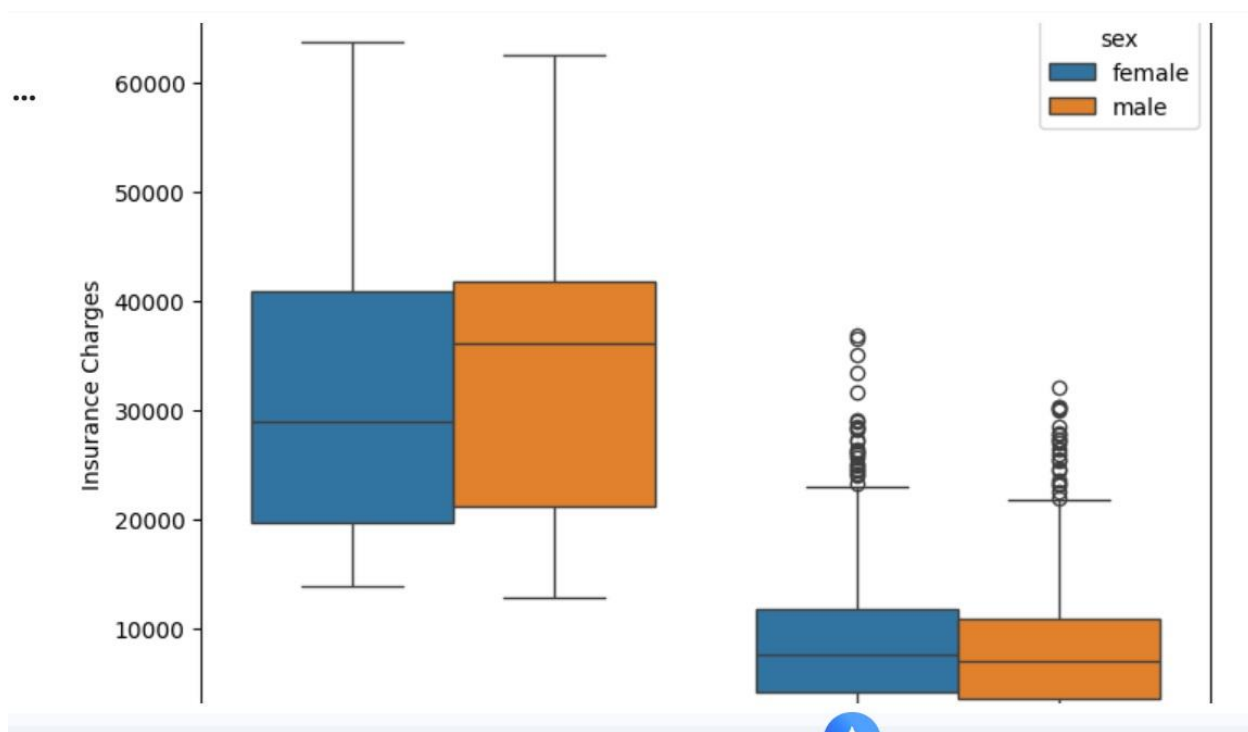
      Minimum Insurance Charges:
      1121.8739

      Average Charges by Smoker Status:
      smoker
      no      8434.268298
      yes     32050.231832
      Name: charges, dtype: float64
```

Q9.







Q10.



```
# ===== Analysis Summary =====  
  
# 1. Average Age and BMI:  
#   - The average age of customers is approximately 39 years.  
#   - The average BMI of customers is approximately 30.66.  
  
# 2. Effect of Smoking on Insurance Charges:  
#   - Smokers have significantly higher insurance charges than non-smokers.  
#   - The average charges for smokers are around ₹32050,  
#     while non-smokers pay around ₹8434.  
  
# 3. Region with the Highest Number of Customers:  
#   - The Southeast region has the highest number of customers  
#     in the insurance dataset.  
  
# 4. Interesting Observations:  
#   - The distribution of charges is right-skewed.  
#   - Most customers have a BMI between 25 and 35.  
#   - The dataset contains more non-smokers than smokers.  
#   - Insurance charges are much higher for smokers, as shown in the boxplots.  
#   - Age and BMI have a weak positive correlation with insurance charges.
```